

Prologue to Pythagorean Renaissance: Ibn Turka's *Investigations* (1420) as Opening Anthem of the Scientific Revolution, Islamic and Otherwise

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a wise and letters play of all you can ceive¹

—JAMES JOYCE

The more a thing is occulted, the more it is made manifest.²

—IBN TURKA

He errs not who says being—substance and essence—is one.³

—GIORDANO BRUNO

The prologue to Ibn Turka's *Investigations* (*al-Mafāḥiṣ*) of 1420—the first summa of Islamic Pythagoreanism and consciously scientifically revolutionary—does not merely prettily excuse and describe its contents. In sharp contrast to its contemporary and later Latin cognates, it *is* a psychedelic mantra and talismanic machine itself.

This Timurid imperial masterwork is the most philosophically systematic formulation of lettrism (Islamic kabbalah) ever penned.⁴ But its prologue (beforeword) goes farther to take the Baroque rule *form is content* radically literally, making its opening lines the most lettrarily per-

¹ *Finnegans Wake*, 237.

² *Sharḥ-i Naẓm al-durr*, 6.

³ *Cause, Principle and Unity*, 90; I have modified de Lucca's translation of Bruno's seminal treatise. Or as paraphrased by Jack Lindsay in the introduction to his own translation, 25: "Reality is thus defined as a unitary process in which matter is both content and form."

⁴ Greek alphanumericism was the late-ancient source of medieval Hebrew kabbalah and Arabic lettrism, pan-Mediterranean twin sisters, both of which sciences became favorite vehicles of early modern Pythagoreanism throughout the Greater West; see Acevedo, *Alphanumeric Cosmology*. Needless to say, I invoke the descriptor Pythagorean (*fīthāghūrī*)—everywhere explicit and implicit in my Persianate sources—as icon of *mathesis* to decolonize the modern historiography of science, still violently colonialist-Orientalist. For comparative context see Joost-Gaugier, *Pythagoras and Renaissance Europe*.

fect Pythagorean apothegm to be expressed in Arabic or any other language, and its larger framing the most poetically perfect Persianate epitome of Hermetic ascent-descent-ascent cosmo-hermeneutics. And since metahistory is structurally often weirdly literary too, *Investigations* as a landmark scientific manifesto served as gorgeous prologue in turn to the post-Mongol, early modern efflorescence of Persianate cultures across the vast Afro-Eurasian ecumene. Predicated on the empiricist mathematization of the cosmos (*mathesis*), Ibn Turka's totalizing project became source code for the Mughal (Indo-Timurid), Safavid and Ottoman Empires and smaller states alike, inspiring everything from cosmology, poetry, architecture and armor to observational astronomy, music, mind-control magic and predictive computing—a seachange I call the Occult-Scientific Revolution.⁵

For comparative purposes, Ibn Turka (1369-1432)—Chief Judge of Isfahan and architect of Timurid scientific imperialism—may thus be profitably read as a Muslim Iamblichus and Cusanus and Pico and Bruno and Dee. Yet he was considerably more influential than all these Christian peers, for a much greater part of the human race, until the colonial period. Simply put, he is the first thinker in the Greater Western (Hellenic-Judeo-Christo-Islamic) tradition to develop a mathematical-theurgic philosophy, very consciously Pythagorean and very consciously Modern, that successfully challenged the hegemonic Aristotelian-Avicennan system to science-shifting, culture-shifting and indeed world-changing effect.

A comparative case in point is Ibn Turka's contemporary and fellow lawyer Nicholas of Cusa (1401-1464), who shared his Pythagoreanizing vision.⁶ But it took nearly five centuries for the German cardinal to be canonized as a Saint of Science, thanks almost exclusively to his reception by Bruno; his own influence, mostly political, was limited to parts of Germany and Italy, and led to the construction of no observatories. By contrast, our Iranian philosopher boasts major scientific and political impact in his own lifetime—including the construction and program of the epochal Samarkand Observatory—and several centuries after, with sweeping intellectual and cultural ramifications from the eastern Mediterranean and East Africa to the Strait of Malacca and the gates of China.

At the same time, as a humanist champion of the Moderns over the Ancients, and as a judge known for his fierce defense of the weak against the powerful, Ibn Turka was subject to three state inquisitions contrived by jealous and vengeful colleagues. His first two defenses were successful but his third was not; after five years of wandering exile he died, impoverished, in legal limbo. This dramatic story arc as persecuted visionary overwhelmingly vindicated after death is classically Brunian, which alone should make him a Saint of Science too in the textbook Scientific Revolution story—and so ideal interlocutor to his fellow radical Pythagoreanizers and indirect heirs Kepler and Bacon, Galileo and Newton. But in a poetically perfect inversion of the modern hagiography of Giordano Bruno (1548-1600), ultimate Martyr for Science, Dr. Littleturk

5 Melvin-Koushki, "An Islamic Scientific Revolution?"

6 See Albertson, *Mathematical Theologies*.

has instead been historiographically burnt at the stake, and *Investigations* remains on the academic Index to this day.⁷

This Brunian-Ibn Turkian dialectic is not just exceptionally useful for decolonizing the violently colonialist twentieth-century “Scientific Revolution” narrative: it therapeutically reminds us mechanized, industrialized, pseudo-Pythagorean, shit-Platonist, AI-golemed, mind-killed moderns that *mathesis* was always meant to marry the True to the Beautiful.

For both men as good kabbalists were pioneers and champions of the Baroque rule: because being is both substance and essence, matter is both form and content, so cosmos is *poiesis* is Magic.⁸ (Ironically, the Nolan loves the Greek anti-Persian slur *magic* because it is specifically Persian, per the biblical Magi; his Isfahani predecessor naturally finds other terms of art more intellectually and politically useful, and never uses *sihr*—usually translated as “magic” when legal and “sorcery” when illegal—as a master category.⁹) Bruno’s crime was not just his radical Pythagoreanism, but the scintillating, intoxicating, dangerous, delicious tang of his prose in both Italian and Latin. Ibn Turka is likewise rightly acclaimed the first author in both Persian and Arabic to marry gorgeous, tricky, prosimetric, psychedelic style to philosophical-scientific argument. Their shared authorial ethos and crime: *text as sorcery as starship*.

To start this long-overdue conversation, and in honor of this volume’s mostly Latin focus, consider Bruno’s opening, totalizing statement in his *De triplici minimo et mensura* of 1591, now celebrated as a foundational source for the Scientific Revolution story, because a beautifully muscular epitome of *mathesis*:

Mens super omnia Deus est. Mens insita omnibus natura. Mens omnia pervadens ratio. Deus dictat et ordinat. Natura exequitur atque facit. Ratio contemplatur et discurrit. Deus est monas omnium numerorum fons, simplicitas omnis magnitudinis et compositionis substantia, et excellentia super omne momentum, innumerabile, immensum. Natura est numerus numerabilis, magnitudo mensurabilis, momentum attingibile. Ratio est numerus numerans, magnitudo mensurans, momentum aestimans.

7 Irony of painful ironies: most of the handful of modern accounts of Ibn Turka still reflexively paint him as a generic Shi’i Sufi mystic—exactly the charges he vehemently defended himself against in his remarkable apologies and creedal statements—and take no notice of the *Mafāḥiṣ*. My 2012 dissertation was the first study to point out the utterly obvious: as the author himself everywhere states, *Investigations* is the centerpiece and culmination of his life’s project.

8 Yates, *Giordano Bruno*; de León-Jones, *Giordano Bruno*; Bruno, “On Magic.” See also Maiorino, *The Cornucopian Mind*; and of course Deleuze, *The Fold*, for a sorcerous reading of fellow Pythagorean Leibniz as soul of the Baroque.

9 See Melvin-Koushki, “The Meanings of Islamic Magic.”

Influit Deus per Naturam in rationem. Ratio attollitur per naturam in Deum. Deus est amor, efficiens, claritas, lux. Natura est amabile, obiectum, ignis et ardor. Ratio est amans, subiectum quoddam, quod a natura accenditur et a Deo illuminatur.¹⁰

My translation—in an Ibn Turkian mode:

Mind above all is God; Mind within all is Nature; Mind moving through all is Thought. God speaks; Nature writes; Thought reads and moves to write again. God is the Monad—source of every Number; the Simple—substance of every Measure and Composition; the overflowing Motive—above all Motion: Innumerable, Immense. Nature is Number numbered, Measure measured, Motion moved. Thought is Number numbering, Measure measuring, Motion moving.

God flows down into Thought through Nature; through Nature Thought ascends back to God. God is Love, Cause, Clarity, Light; Nature is Beloved, Object, Fire, Burning; Thought is Lover, Subject, Inflamed by Nature, Illumined by God.

That only minor adjustments were necessary to make Bruno's Latin style express Ibn Turka's Arabic philosophy proves their wonderful scientific resonance, despite the almost two-century age gap and Christian-Muslim political divide. Their respective Italian and Persian styles are even more ornately resonant. To further retranslate the former, per the opening epigraph: *He errs not who says Pythagoreanism—Iranian and Italian—is one.*

In *Investigations*, however, and throughout his gorgeous Perso-Arabic oeuvre, the Isfahani takes the Nolan's now-celebrated *cabala-teologia-filosofia* triad further, both stylistically and scientifically—naturally the same thing for lettrists.¹¹ His ontogrammatology rhymes beautifully with the cosmic semiosis of Copernicus-to-Kepler, and especially Kepler's radically Pythagorean *mannerist* project.¹² It is no accident that Ibn Turka and Kepler are the greatest writers and readers of the two most mathematically accurate star-tables ever produced before telescopic observation, Ulugh Beg's and Tycho Brahe's respectively: the Baroque in Starstraining mode.¹³

But it is only in *Investigations*' prologue that form fully becomes content. This is thanks to the more perfectly alphanumeric, algorithmic and hence cosmic nature of Arabic, in contrast to Greek, and especially Latin, whose alphanumeric-algorithmic virtues are strictly partial, so by definition less celestially attuned. By the same token, the miraculously inimitable language of the Quran—especially the mysterious letter series (*muqatta'āt*) opening certain suras, the well-spring of lettrism—had long since become synonymous in various genres with astonishing

¹⁰ This passage is transcribed and studied in Bönker-Vallon, "The Measurement of the Immeasurable."

¹¹ More specifically, Bruno's self-proclaimed project is "a cabala of theological philosophy, a philosophy of cabalistic theology, a theology of philosophical cabala"; see Yates, *Giordano Bruno*, 262; de León-Jones, *Giordano Bruno*, 24.

¹² Hallyn, *The Poetic Structure*.

¹³ Melvin-Koushki, "Sultan-Scientists."

Joycean feats of wordplay, most impossible in Latin, and some with talismanic potency. A distinct literary-scholarly genre with highly codified rules, the structure of the Arabic prologue in particular descends from pre-Islamic poetry as gorgeous and therefore very real (but legal!) sorcery.¹⁴

More generally, pre-Mongol Arabic philosophical prose could be formally elegant, and pre-Mongol Persian poetry (especially Nizāmī's, often occultist) could incorporate scientific content. But it is only with the Great Weiriding that was the Mongol conquest that star science and letter science began to boom in tandem—to the point that Ibn Sīnā (Avicenna) himself was pseudepigraphically rewritten as a bestselling *astrolettrist* grimoirist. Ibn Turka subversively embraces this transmogrification of the Second Aristotle by marrying him to the Second Plato, Ibn 'Arabī (1165-1240), Andalusian Supreme Master, whose oceanic prose is frequently mindbendingly theurgically ornate. Crucially, he is the first to add Sa'd al-Dīn Ḥamūya (d. 1252), the Andalusian's Iranian twin, as co-father of lettrism, whose jazz-philosophical prose goes farther to achieve *ontological ornateness*.¹⁵ (That Ḥamūya's son played a central role in the epochal Mongol conversion to Islam through the gift of his father's talisman and robe is metahistorically poetically perfect.¹⁶) So inspired, their Isfahani heir indeed became the first author to sorcerously marry Arabic (and Persian) Style to Science.

Speaking of Magic: even pre-Mongol Arabic grimoires focused on the invocation of planets (talismanry) feature no special ornateness, apart from an often heightened register for prayers themselves; and their prologues are mostly workmanlike, as befits technical manuals. Yet they are still quite different in tone and style from their Latin translations, cognates and descendants as a rule.

The classic comparative example is the tenth- or eleventh-century *Goal of the Sage* (*Ghāyat al-ḥakīm*), produced under Pythagoreanizing Brethren of Purity auspices imported to Islamic Iberia—with which shadowy group of court occultists Ibn Turka and his colleagues so powerfully identified as to call themselves by the same name.¹⁷ For reasons Eurocentric, modern historiography has made its thirteenth-century Castilian Spanish then Latin translation, somewhat garbled, truncated and ideologically altered, under the also garbled mystery title *Picatrix*, the ultimate symbol of Arabic-to-Latin Magic—based on its comparatively slim, rather diffident and often hostile Latin Christian reception only. This has led the last several generations of scholars to labor under the assumption—facile and entirely false—that this makes the Arabic original central to the story of Islamic Magic too. To the contrary, other Arabic and Persian grimoires were at least equally popular, and many are more attested in the surviving manuscript record, despite all depredations, by an order of magnitude. But since none of these bestsellers

14 Selove, "Magic as Poetry."

15 Uy, *Lost in a Sea of Letters*; idem, "Riffing on/as the Real."

16 Uy, *Lost in a Sea of Letters*, 6.

17 Melvin-Koushki, "The New Brethren of Purity." The encyclopedic *Rasā'il* of the original Ikhwān al-Ṣafā', also synthesizing and popularizing, is likewise founded on the premise of number as the root of all philosophy and science; but they cannot be described as ontologically ornate in the New Brethren mode.

were translated by a European hand, even badly, presto-chango they do not exist.¹⁸ This Arabic-cum-Latin grimoire-as-symbol thus serves as a fitting prologue to the European Renaissance fiction, now Euro-American colonial fact, of the passing of the scientific torch from Greek Paganism to Latin Christianity (not Greek) with a perverting detour through Islam, that simultaneously pseudo-Hellenic and pseudo-Abrahamic Neopagan aberration fit for stripmining at best.

The stylistic and sociopolitical divergence between the prologue of the *Ghāya* and that of the *Picatrix* is therefore decolonially instructive. The unadorned Latin comes off as painfully effusively theological for apologetic purposes; the Arabic is not just shorter and tighter, but serenely unapologetic. It is also explicitly democratizing, since the grimoire synthesizes well over 200 sources for the convenience of the modern practitioner (both in the tenth century and the twenty-first). Early modern Persian bestsellers follow the same democratizing, encyclopedist approach, synthesizing dozens to hundreds of other occult-scientific works.

Ibn Turka's prologue to the *Mafāḥiṣ* then marks a seachange: it is equally unapologetic, but masterfully ontologically ornate, so thoroughgoingly elitist—yet paradoxically more democratizing too. The book itself cites only the mindbending Ibn 'Arabī and Ḥamūya (and the proto-lettrist alchemist Jābir ibn Ḥayyān) as sources, while insisting on the unprecedentedness of its own theoretical framework and cosmological diagrams.¹⁹ But it also bends over backward to couch that new framework in the technical terminology of all contemporary schools of thought to ensure the widest scholarly uptake, and the diagrams are unusually simple and fully explained, with a comprehensive two-page table as key.²⁰ Following the same radically ecumenical strategy, Ibn Turka's more gorgeous Persian treatises, written to transform warlords into philosopher-kings and Avicennan philosophers into Pythagorean lettrists, are also more accessible.

That these paradoxical virtues made his Arabic masterpiece perfectly untranslatable into Latin is proof of comparative concept. That my English translation of its prologue below is illegible as Science to modern specialists without this, my own prologue—apologetic and non-ornate in Latinate fashion—likewise speaks volumes.

Form is content. The Baroque motto turns out to be a weirdly accurate English translation of the New Brethren's quranic mantra: *He is the First and the Last, the Manifest and the Occult* (Q 57:3). The wonderfully Baroque rules of Arabic (and New Persian) epistolography, prosody and all other literary genres had long since been worked out, whereby even slight deviations from the stylistic norm became pregnant with meaning. Indeed, the mere subtraction or reordering of a formal element in a diplomatic letter could incite war.²¹ But the New Brethren were out to Hermetically marry all opposites and Heaven to Earth through language, whether cosmic or bu-

18 Dan Attrell and David Porreca's introduction to their masterful translation happily avoids this facile assumption, emphasizing instead the *Picatrix*'s underwhelming reception history.

19 As with the *Picatrix*, modern scholarship has naturally focused far more on Jābir's Latin reception, and especially the pseudo-Geber corpus, than any of the original Arabic. See e.g. Newman, *The Summa perfectionis*.

20 While Ibn Turka follows this synthesizing procedure throughout his oeuvre, in his important treatise *On the Splitting of the Moon and the Last Hour* he arranges all schools of thought into a hierarchical seven, from traditionalist literalism at the bottom to lettrism at the top as apex of human perfection; see below.

reaucratic, poetic or military. Per Ibn Turka's opening epigraph: *The more a thing is occulted, the more it is made manifest.*

A case in point Ḥusayn Vā'iz Kāshifī (1436-1505), Timurid imperial preacher and judge, fellow New Brother and popularizing Persian astrolettrist author par excellence: the "Agent of the Unveiler" perfectly threaded that Occult-Manifest needle to create an Everyman's Library to edify and delight elite and commoner alike. (So perfectly, in fact, that his gorgeous, bestselling Persian adaptations of Arabic classics were used as language-training manuals for colonial operatives in British India, and continue to be popular today; and one of his five grimoires has far out-sold and continues to outsell the *Picatrix*.)²² The opening to his *Treasury of Epistolography* (*Makhzan al-inshā'*) is therefore especially salient. No mere chancellery manual—rather a stylistic LLM *avant la lettre*—, this visually gorgeous, encyclopedic, unprecedentedly deconstructionist imperial-bureaucratic matrix empowers all state functionaries to algorithmically achieve perfect prosimetric elegance in their own right, thereby rewriting the Court and the State as a Second Book—and its very first line is exactly this mantra.²³ It must be emphasized that so abrupt an opening is designed to be stylistically shocking, and makes the mantra *neon*. But here Kāshifī is just faithfully, ornately reprising Ibn Turka's own ornate political science, whereby the Timurid Empire must be read and beautifully rewritten in astrolettrist terms: Science and Art as Occult, Bureaucracy and Military as Manifest, to imperially create (per Newton's landmark English translation of the Emerald Tablet) "ye miracles of one only thing."²⁴

An imperial engine, *Investigations* is Science and Art as *one only thing*. The book's prologue is accordingly a neon performance and unprecedented meta-performance of the basic Arabo-Persian literary device of *barā'at al-istihlāl*: a virtuoso opening brilliantly, beautifully telegraphing its text's theme as a whole. True to lettrary form, Ibn Turka teaches the ensorcelled reader to take all such rules cosmically literally, the better to marry Heaven to Earth. For this device's literal meaning: *the gorgeous sighting of the Crescent Moon*.

The Avicennan (and Suhurvardian) Moon is the Active Intellect, base camp and High Castle from which philosophers and prophets explore the structure of the cosmos. But for Ibn Turka, radical Sunni imamophile, Imam 'Alī as inventor of Letter Science is Pythagoras is Moon is Panopticon is Mathematical Wormhole to Reality. Hence his prologue's mathematizing-poeti-

21 Melvin-Koushki, "The Delicate Art." Fun fact: my first publication, this 2011 article is based on a 2008 graduate seminar paper written just after I had discovered Ibn Turka as a master ornate prose stylist, but was still wholly ignorant of the revolutionary nature of his Pythagoreanizing lettrist project, much less the centrality of its themes to history of science.

22 Subtelny, "Kāshifī's *Asrār-i qāsimī*."

23 On this imperial matrix and its strikingly deconstructionist algorithm see Mitchell, "To Preserve and Protect" The *Treasury's* eager reception at all early modern Persianate imperial courts (Safavid, Mughal, Ottoman, Uzbek) and even regional ones is eloquent testament to cultural sociopolitical and philosophical continuity despite military rivalry. Several of the best and earliest surviving copies, for example, were preserved in Ottoman collections and are now available online. Especially visually lovely is Manisa, MS 2768, probably copied within a few decades of Kāshifī's death: <https://portal.yek.gov.tr/works/detail/98333>.

24 Melvin-Koushki, "Imperial Talismanic Love"; Newman, *Newton the Alchemist*, 145.

cizing theurgic declaration *Letters are Moons*—whose quranic miraculous splitting (*shaqq-i qamar*) is the splitting of the Pen that writes and gorgeously rewrites the Algorithmic world. The story of the “Scientific Revolution” is likewise peculiarly Selenocentric, despite the aggressive Heliocentrism-as-One True Pythagoreanism orthodox Eurocentrist branding, what with Kepler’s extrapolations based on Galileo’s telescoping of Earth and Jovian Moons. Yet his cosmic mannerism, though conceptually intensely Pythagorean and equally astrological, is neither stylistically ornate nor explicitly ritually theurgic—aka *mind-telescopic*—in sharp contrast to Ibn Turka’s.²⁵

So far Science; what of Art? *Investigations*’s prologue, that perfect because perfectly literal incarnation of *barāʿat al-istihlāl* as Moonsighting as Moonsplitting, shockingly openly transgresses the strict formal code governing the genre in another act of positive sorcery. Not only is the opening prayer to God and Prophet unprecedentedly lettrarily virtuosic; the rest of the prologue also becomes psychedelic by flaunting every last topoi to sublimate them all.

Ibn Turka as a Modern author is supremely self-assured. Here as elsewhere he identifies the occasion of this work as the exigency of Time itself, more than the standard importuning of a merely human patron, student or friend, making it an astrological inside joke. He provocatively refuses to perform the standard self-deprecation of author-as-wretch—just as he would refuse to bend the knee during his three inquisitions, despite the danger and punishing consequences. Even more provocatively, he dismisses all sources save for two of the most notorious, thanks to the inspiration of his infamous Kurdish lettrist-alchemist-geomancer mentor; and declines to give a book plan.²⁶ Most shocking is the fact that his prologue offers no closing prayer.

This stylistic subversiveness is of course a Baroque trick. The prologue and book itself is the prayer and textual embodiment of the New Brethren mantra, which proves Time to be a manipulable mathematical-linguistic construct. Its book plan is similarly encoded in the prologue’s Pearl-Peach-Planet cosmic trinity, whose Three Globes of Light correspond to the Written-Spoken-Mental forms of the mathematical Letter in purely poetically Persianate guise. The *Mafāḥiṣ*’s meta-performance of authorial humility as Descent between two Ascents—its glorification of the Written over the Spoken, Sight over Hearing—is likewise gorgeously Hermetic.²⁷

25 On this comparative theme generally see Melvin-Koushki, “Selenocentrism and Heliocentrism”; on Ibn Turka’s gorgeous and unprecedented Persian treatise by the same name see *ibid.*, 242-44. It must be emphasized that lettrism—the endless splitting of language to release mathematical consciousness, like spirit from body—is simply the most literal possible emulation of the Cut-Up Letters (*muqattaʿāt*) opening 29 suras, a mindbending feature not yet found in the Hebrew Bible or Greek New Testament. For Ibn Turka, the continuum Lettercutting to Moonsplitting to the Twohorned Sword of ʿAlī as *Pythagoras redivivus* and inventor of letter science could not be more patent. Cf. William S. Burroughs’s infamous Cut-Up Method, likewise designed to prove his trickster thesis that *language is a virus from outer space*.

26 That mentor is Sayyid Ḥusayn Akhlāṭī (d. 1397), Cairene pivot of the New Brethren network, radiating to Iran to India to Anatolia; see Melvin-Koushki, “The New Brethren.”

27 On Ibn Turka’s new sensory theory, which boldly overturns the Aristotelian-Avicennan hierarchy, see Melvin-Koushki, “Vehicle of Total Perception.”

Or rather, Planet-Pearl-Peach: the surprising structure of the book is the Hermetic journey epitomized. It first flies up to the realm Mental (Planet), the metaphysical ceiling of being, constellated by timemaking bodies of Light; then dives down to the opposite ontological extremity that is the realm Written (Pearl), to find the same Light in the sheerest blackness that is physicality as Ink; then enters the Paradise of the realm Spoken (Peach), there to pluck and savor its Sunkissed Luniform Fruit. And oh how juicy and eternal-lifegiving that Fruit: the fourth and final part therefore discourses at greater length on the Letter itself in terms of Form as sorcerous Content.²⁸

All three Globes are metaphorical (mindtransporting) Moons, face of and portal to the Glowing Timeless Beloved at each of the three levels of being.²⁹ The fourth part secures the marriage of Above to Below in the most cosmo-stylistic of terms to encourage the imperial cultivation of Paradise on Earth. So does this scientific-poetic reordering of the Chain of Being remain the Tetractys: Unit Numbers as Necklace as Jewelstrung Rosary as Highway to Heaven—and back to Earth to make it too a Globespangled Garden. (That *Investigations's* central diagram, the Ṭahawī Cycle, is the first Modern lettrist version of the Tetractys is a case in point.) And so does it subvert the classic Persianate poetical tropes and their Sufi associations to rewrite Plato's Cave itself as Mine of Mind and Pearlbed of Being and so Underground Gateway and Prologue to the Stars.

Our Isfahani philosopher's subversion of Science and Art by Hermetically recoding Plato's Cave was metaphysically and physically successful for centuries—the ones we presume to call early modern. Metaphysically, the *Mafāḥiṣ* was a major source for Mīr Dāmād (d. 1631), fellow Isfahani architect of the successor Safavid state and Kepler's contemporary: despite being the most important Western philosopher of Time (*metatime*) of the early modern era, he too has been almost totally erased from the historiographical record for the stylistic crime of Pythagorean-lettrist ontological ornateness. Physically, their combined philosophical-lettrary firepower fueled the astonishing transformation of Safavid New Isfahan into the world's first purpose-built *Pythagoropolis*, Solomonian-Hermetic *city-as-talisman-and-observatory* and so Islamic New Atlantis.³⁰

28 The unspecified Fruit here referenced is most likely Orange, according to classic Persian poetic usage; I translate it as Peach to better capture the amatory charge and for the sake of alliteration. Note that Ibn Turka's identification of East with Peach, West with Pearl and Up with Planet naturally encodes a geographical riddle: from Isfahan to the east were the arid inland forest cities famed for their fruit production; to Isfahan's immediate west is the Persian Gulf, iconic center of the pearldiving industry from time immemorial; and the arid Iranian-Transoxanian upland clime—the stars shining brilliantly above—perfect for an observatory.

29 A further key to this riddle is 'Alī's status as Moon to Muḥammad's Sun for Shi'i and Sunni alike, and therefore both imperial Beloved and cosmic Gateway, per the ubiquitous Prophetic hadith: *I am the city of knowledge and 'Alī is its gate*. Similarly resonant is the standard Persian word for prologue, *dībācha*, an early Persian-to-Arabic-back-to-Persian term expanding the original sense of silk brocade to embellishment in general, making a beautified *face*, a beautified *garden* and a beautified *preface* ontologically equivalent in early modern Persianate usage.

30 Melvin-Koushki, "Mīr-Dāmād's *On Doffing*"; Emami, *Isfahan*.

Metahistorically also poetically perfect is New Isfahan's rhyming with contemporary Naples, or rather Neapolis, that New City with ancient Pythagorean bones, where Bruno the Nolan was formed. But where Bruno's Naples was styled New Jerusalem and Sun City of the Spanish Empire at its high noon,³¹ Ibn Turka's hometown and resting place became under Safavid imperial auspices the City of the Moon ('Alī), whose Islamic higher octave is Mercury (Hermes), with modern Pythagorean bones—and so a modern Starship to boot.

Talismanic city-planning aside, the most iconic apotheoses of Timurid imperial-theurgic architecture are two. The Indo-Timurid Tāj Maḥal—that glorious mausoleum marrying Sun to Moon for Underworld to Overworld journeying—is now a tourist trap, so no longer a legible machine for achieving out-of-body philosophy, much less a computer for making Paradise on Earth. Thanks to the modern colonialist-Salafist use of “Safavid” as a slur and endless Euro-American sanctions and wars, the Shaykh Luṭf Allāh Mosque on New Isfahan's Map of the World Square is less of a trap for international tourists. But its imperial Starship design, overwhelmingly, psychedelically gorgeous, is the most architecturally and visually literal possible incarnation of *Investigations'* Planet-Pearl-Peach series ever built. This is despite and because of its riddling reversion to Pearl-Peach-Planet: white-on-azure Letter ascends to rampant-tendriled Fruit ascends to celestial-mathematical black body of Light. The acoustics for divine-name mantra recitation are gobsmacking too. But the monumental quranic calligraphy and blazing lines of Persian poetry are quite explicit that Hearing is ontologically superior and therefore epistemologically inferior to Sight.

Because Art is Science and Time is Tricksy, Persianate book painting also suddenly boomed during the Timurid period; its frequent psychedelicality gave birth to a new Safavid literary genre, the *album preface*, which explicitly embodied Ibn Turka's ontogrammatology.³² By the same token, gorgeous paintings of the Prophet's Ascent to the Ceiling of Being (*mi'rāj*) began to figure as manuscript frontispieces within a few years of our Timurid philosopher's death, and became a standard Safavid, Mughal and Ottoman visual idiom for the beautification and imperialization of Plato's Escape and Hermes's Return.³³

Book-Building-Being remains the opening anthem of the “Scientific Revolution” story (and the new-old AI story too, however enshittified). Ontogrammatological apotheosis of such stories, *Investigations'* prologue's opening lines are likewise beautifully, quranically anthemic—but go further to embody their very source code in a daring lettrary feat: rewriting the sura-opening letter series of the Quran itself as Second Book and cosmic-mathematical text.

To that revolutionary scientific-poetic end, Ibn Turka plays on three Arabic letter trinities he puts at the heart of his Pythagoreanizing project: عدد عود عبد (*DD WD BD*). The most basic English paraphrase of these rhyming verbal roots: *to number is to return is to serve*. Since the

31 Marino, *Becoming Neapolitan*. That Isfahan and Naples both hit a population of around 200,000 at the same turn of the seventeenth century moment, making them major cities, likewise rhymes.

32 Melvin-Koushki, “Of Islamic Grammatology.”

33 Art historians have investigated this florescence at great length; see e.g. Gruber, “The *Mi'rājnāma's* Afterlife.”

ground of Arabic ornateness—in sharp contrast to Latin—is the organic proliferation of word branchings from every primordial root, each does heavy epistemological and ontological lifting. “Number” encodes “Training” through the ultimate “Instrument.” “Return” (to the One) encodes the Pythagorean “Lute” as divine-names “Mantra” and planetary “Prayer” as means of Ascent to, because “Instrument” of, the Music of the Spheres, to talismanically draw its potency down (the Return again). “Servant” reconceptualizes the Descended human condition as soul of cosmic “Alienation” best remediated through ontological “Ornateness”—and direct visual Observation mathematically *precisely* Calculated. The same root (*‘abd* = *bad’* < *badī*) graphically privileges Ibn Turka’s favorite quranic proof-text for the marriage of Science and Art as the central human mission (Q 38:45): imperial lettrism as Moon Science is peculiar to *men of main and vision*.³⁴

Constructing the cosmos as World Tree, spanning from the watery subterranean realm to the stars, and reconceiving Earth as Paradise, this prologue’s wordplay is not just ontologically ornate, but quite literally *spectacular*. That it coincided exactly with the construction of the Samarkand Observatory (1420) and the inception of its epochal star-tables—so unprecedentedly mathematically precise they were used everywhere from the Strait of Malacca to London to the nineteenth century—is likewise metahistorically poetically perfect.

True, like Bruno, Ibn Turka set himself up for historiographical Inquisition by subordinating all contemporary schools of thought and practice to a Pythagorean-lettrist code; its hierarchical elitism yet radical ecumenism was an easy target for gross misrepresentation at court. It remains so, being irreducibly *Islamic*. To the untutored eye, his Science all too easily reads as Religion at best and Magic at worst: hence the total erasure of the *Mafāḥiṣ* in the last century of scholarship, and the scrubbing of its now-martyred author from the history of science. How, for example, can we moderns read the Sufi’s subjective distress at distance from the divine presence as an expression of the empirical astronomical fact of planetary apogee, or the theologian’s insistence on the literal Splitting of the Moon as entree to language and consciousness itself?

But the Judge of Isfahan asks of us historians justice. As this wretch’s semi-Joycean translation below can only intimate, his call to Calculate (*ḥisāb*) the Code (*bayyināt*) of the cosmos could not be a more clarion epitome of early modern *mathesis*, nor its *barā’at al-istihlāl* a more beautiful Moonsighting as Moonsplitting as modernity’s own prologue.³⁵

34 Melvin-Koushki, “Elevation of Sight.” As Ibn Turka schematizes in his Persian treatise (or rather anthem) *On Letters*: “Three loci of self-manifestation have been created for the letterform through which it manifests and reveals the end and the essence of every thing. The first is Sight, to which the ‘ (*‘ayn*, Eye) in the word *‘abd* refers; the second is Heart [or Mind], to which the *B* in *‘abd* refers; the third is Hearing, to which the *D* in *‘abd* refers.” Thus the Pearl-Planet-Peach progression; and thus the equation of being human (*‘abd*) with divine creation (*bad’*) with literary device (*badī*).

And per his quranic proof-text as to the marriage of Eye to Hand: it is no accident that the atelier of Ibn Turka’s first Timurid patron, Iskandar Sulṭān, is celebrated to this day for combining star science, poetry and book painting in its gorgeous output; or that his second, Bāysunghur, was an accomplished calligrapher in his own right, and likely commissioner of the incomparably monumental Bāysunghur Quran, surviving folios of which are dispersed in collections worldwide, and small fragments of which continue to be sold for eyewatering sums (up to £500,000) by Sotheby’s.



Plate 1: *Mafāhiṣ* opening with author's autograph,
Tehran, Majlis Library, MS 10196, f. 52b

35 Seemingly generic, the term *bayyināt* is also (of course) a technical letterist one (see Q 3:184, 16:44, 35:25): when performing *taksīr* (cognate to Hebrew *temurah*), which involves the *muqattaʿāt*- and Moonsplitting-inspired separation of the letters of a name or word and the writing out of the letternames in full, then the elimination of repeated letters, the term *zubur* refers to the first letters in the full letternames (e.g., the *A* in *ALF*) and *bayyināt* to the remaining letters (*LF* in *ALF*)—so by definition the occult code behind every manifest word, and therefore the world itself.

الحمد لله الذي أعدّ عديد عبادته لفهم كمال مراده بإنزال أعداد يتنات الكتاب وجعله العدة المعتد بها لمعرفة معاده يوم الحساب وأبان بمعدودية أيامه ميعاد أعياد قربه ونفي بعاده على نهج الصواب ويّين بها مدارج عبيده فيه بحسب ما لها من الإحساب والصلوة على محمد خير من أوتي الحكمة وفصل الخطاب وعلى آله وصحبه هداة شوارع الإيقان القائدين إلى مناهل أورد أولي الألباب.

*al-ḥamdu li-llāhi lladhī a‘adda ‘adīda ‘ibādi-hi li-fahmi kamālī murādi-hi bi-inzālī a‘dādi bayy-
ināti l-kitābi wa-ja‘ala-hu l-‘uddata al-mu‘adda bi-hā li-ma‘rifati ma‘ādī-hi yawma l-ḥisābi wa-
abāna bi-ma‘dūdīyyati ayyāmi-hī mī‘āda a‘yādi qurbi-hi wa-naḥyi bi‘ādī-hi ‘alā nahji l-ṣawābi wa-
bayyana bi-hā madārīja ‘abīdi-hi fī-hi bi-ḥasabi mā la-hā mina l-iḥsābi wa-l-ṣalātu ‘alā Muḥam-
madīn khayra man ūtiya l-ḥikmata wa-faṣla l-khiṭābi wa-‘alā āli-hi wa-ṣaḥbi-hi hudāti shawāri‘i l-
iqāni l-qā‘idīna ilā manāhili awrādi ūlī l-albābi.*³⁷

Praise be to God Who has Mathreadied His Numberservants to know His Total Will by Descend-
ing the Numbercodes of The Book and making it the master Calculator for kenning the Return
to Him on the Day of Calculation, distinguishing the Daynumbers for Timenumbered Feastday
betidings of the drawing near [that is His Perigee], negating the distance [of His Apogee]—the
correct Method [giving the correct Sum]—thereby enumerating His servants’ [celestial] degrees
of Ascent according to the capacity of each to make Account; and blessings be to Muḥammad,
greatest among those given Philosophical Wisdom and the Dividing Address, and to his House
and his Companions, guides on [Empirical] Demonstration’s desert paths and conductors to the
quenching pools that are [Mathword] Mantras for the Minded.³⁸



36 Tehran, Majlis Library, MS 10196, ff. 52a-56a, copied and added to between 1420-25, as part of the earliest and al-
most complete collection of Ibn Turka’s works, together with some court correspondence. This was almost cer-
tainly by the hand of Ibn Turka’s best friend and greatest student, Sharaf al-Dīn ‘Alī Yazdī (d. 1454), Timurid im-
perial historian and mathematician par excellence, and the first to make the science of history fully formally
astrolettrist; naturally, he also worked at the Samarkand Observatory for a time. He is singled out for love in
this prologue.

37 The three letter trinities *DD WD BD* are bolded, and plays on *ḥisāb* and *bayyināt* underlined.

38 Needless to say, the plays on direct and indirect quranic references are here so dense as to be unflaggable for
nonspecialists. But as an example, for Ibn Turka the frequent quranic refrain *those possessed of minds* (*ulū l-al-
bāb*) refers to Number as the highest key to reality—hence the talismanic harnessing of quranic divine-name
mantras (*awrād*, also meaning watering holes) through mathematically precise recitation as a central lettrist
technology.

The divine or metaphysical sciences—despite all their methodological varieties and programmatic differences—are by definition the highest objective to which human ambition can aspire and the goal on which the chargers that are striving natures are trained. But it is only a science that admits of not the slightest insinuation of doubt that can definitively show which of the finest riders competing on these sciences' racing grounds has won the palm. To that science lead the piercing Stars that are the Descended Rites: Rising from the horizon of eloquent and gorgeous linguistic Expression, whose Letters are Moons, they efflux forth from the Heaven of their glorious Height the Hearts of all Wayfarers in the Night—yet also manifest from the Earth of their fertile Depth the occult Properties of all that Come to Be.

I've Nightread Moonfortunes of many a folk
but never Seen a Star so happy as that Ruling Benefic!³⁹

It is likewise the singular science that God has spread out as Groundcloth in the abode of His Islam for the Feast of His Speech: He first delectates His servants with those Arts that spring forth from the Earth of their [adolescent] Readiness at the Banquets (*walā'im*) that are His *L*-and-*M*, then feeds them to full Maturity with the Dovemeats (*ḥamā'im*) of His *H*-and-*M*, then gives them to drink of the paradisiacal Water of Eternal Life in the Cups (*tāsāt*) of His *T*-and-*S*.⁴⁰

How constantly have the most vigorous and consummately learned of our community's leading lights therefore sought to acquire this supreme science. Driven by the ravenous cravings of their aspiration, they refused to settle for the stale jerky that is the preserved teachings of all who have gone before, but rather strove after the freshest, juiciest Truths—a quest common to all ages unto eternity. In this quest, some made for the East of Expansion and Manifestation, and succeeded in picking ripe Fruits from the Upper branches of the tree of this science as manifested through Speech; they limited their diet to this and sought nothing more. Others made for the West of Contraction and Occultation, and succeeded in amassing priceless Pearls from the Underwater hoards of this science as manifested through Writing—and by my life, it is they who inherit the special features of the holy Seal of Prophecy himself and what he endowed to his heirs: both the Eternal Speech *taught him by one terrible in power, very strong, who stood poised* by way of his noble forebears, and the Temporal Book he read from the Cosmic Tablet of *He inspired in His servant what He inspired* as embodiment of the rare and ripe fruits unique to the Seal's Garden.⁴¹ God greatly reward [both camps] to the extent that any teacher can be by taught.

39 Ṭarafa ibn al-'Abd (fl. 6th c.), qasida: *qifī wadda'ī-nā l-yawma yā bnata Mālīki*. Per his normal multi-register procedure, Ibn Turka makes this famous pre-Islamic line, richly evocative of desert starguided travel, the opening anthem of his astrolettrist theme. A more literal meaning: "I've seen Sa'ds of many nations / but never a *sa'd* like Sa'd son of Mālīk." Here the play is on *sa'd* as auspicious planet (Jupiter and Venus) in the Ascendant or the four so-named mansions of the Moon. Its invocation here is yet again literally *barā' at al-istihlāl* on several levels—including its allusion to Sa'd al-Dīn Ḥamūya as one of Ibn Turka's two stated sources.

40 Three iconic instances of the quranic *muqatta'āt* as matrix of lettrism, here used for lettrary effect: six suras open with *ALM*, seven with *ḤM* and one with *ṬS*.

However, by restricting their quest for divine Effluxion to these Two directions, [East and West, Spoken and Written]—both equally integral and noble parts of the Path—they neglected to pursue the Third Way, [Up: celestial-mathematical Transcendence]. Rarest and choicest, that Way alone validates and empowers [the first Two] by opening the Gate of Truth (*ṣawāb* = *ṢWAB* = 99), behind which are the well-protected treasures of the Seal's power and intimacy with the Divine.⁴² They contain many collected treasures, and above all Necklaces of priceless Jewels, with which [the Seal] provisioned his [True] Son, Seal of Sacral Power and Standard-bearer of Science and Guidance.⁴³ And these Decade-Knotted Necklaces are *Number*—the spiritual-noetic form of the Book as Descended from the supernal Pen upon the noble Tablet.⁴⁴

For it is Number that is the supreme means of mastering infinitely numerous sciences and all most profitable. It is the primordial Mine whence come the gems preserved in all the standard and mainstream scholarly disciplines, whose formal and technical terminology developed by scholars over generations has veiled them with conventional terms of art that *God has given no authority*.⁴⁵

Quranic verses showing the majesty of Number are naturally beyond counting, but—[like all ways of knowing and doing in the world]—are divisible into two types: those sent Down from Heaven and so passively recited, and those sprung Up from Earth and so actively reciting.

The first type you passively experience as a Descriptor that manifests the Divine gift in an absolute sense, per the Most High's statement: *A recompense from your Lord, a gift, an accounting*.⁴⁶ As such, it is only through Counting that the gates of paradise can be opened, per the Prophet's saying: *God has ninety-nine Names—one hundred save one—and whoso counts them shall enter Paradise*. (Note his emphasis: *one hundred save one*.)⁴⁷ This is the clear Pool and paradisiacal River of Multiplication whose infinitely abundant Water the Seal serves directly to all his intimates, regardless of hailing, who are liverparched and heartsick [for True Science]—and

41 Q 53:5-6 and Q 53:10: these verses refer to Gabriel as Hermetic Messenger to the Prophet. Here Ibn Turka both honors and respectfully critiques Ibn 'Arabī, his other main source, who famously declared *West is best!*

42 A reference here is Q 36:14: *When We sent unto them two men, but they cried them lies, so We sent a third as reinforcement*. The number 99 refers to the technological harnessing of the divine names as mantras; see the opening and below.

43 Technically Muḥammad's son-in-law and cousin, but here his sole Heir: 'Alī ibn Abī Ṭālib, fourth Sunni caliph and first Shi'i Imam. (Ibn Turka's given name is 'Alī.) Note that "'Alī" translates to "exalted"—here in a technically *celestial* sense.

44 This play on '*uqūd*—meaning both knottings in necklaces and numerical decades, and here infused with Jewels and Pearls—reprises the general theme of base-ten planetary prayer as Cosmic Rosary as Tetractys.

45 Q 12:40, 53:23.

46 Q 78:36. Marginal lettrist equation: that this "gift" ('*aṭā*, 'ṬA) is ontologically "Numbers" ('*a'dād*, A'DAD) is signified by their shared value of 80.

47 See e.g. Bukhārī, *Ṣaḥīḥ*, K. *al-Tawḥīd* (98), 12, no. 7392. Marginal lettrist equation: 'Alī ('LY) = 110 = *alif* (ALF). Wonderful numberplay: 'Alī is that "save one" as cosmic avatar of the letter A, Pen of Creation, the number One—yet 110 gematrically reduces to Two, the value of B and the opening letter of the Quran (and the Torah), per his famous declaration: *I am the Dot under the B*. In other words: 'Alī as Pythagoras reborn is the Occult Uncity within the Duality of all Creation. See my forthcoming translation of Ibn Turka's treatise *On the Dot*.

served in Cups of Silver, by the hand of ‘Alī, of his own House.⁴⁸ Whoso drinks of that Water will never thirst again.

How great the gulf between him and the clouds:
they bestow drops; he, a flood.⁴⁹

These virtues are no mere double draught of milk.⁵⁰

The second type you actively experience rather as [Isrāfīl’s Time-Ending] Horn, in which all temporal forms are Gathered Up.⁵¹ For you well know the enlightening Rites of the Law and the Prophetic Institutions are rooted in their rigorous manifestations through Decade-Knotted Necklaces of Number, whose forms, revelatory of the Essence of the All, have been so exactly cast [as to serve too as Talismanic Rings]. As such, were the ritual prayer, alms tax or any other holy rite to be stripped of their Number Knots and Gems so well known to the public, not the slightest trace of them would have ever been seen or known or reported. More: any investigation of all the specialized arts and civil professions and all the customary norms of expression and traffic—upon which all markets are predicated and through which all human organizations and assemblies, public or hermetic, without exception cohere—would reveal the whole to be merely formal variations on and expansions of the same rule, embodying its myriad iterations. For there exists no scholarly grouping or proper assemblage of any kind but it is decked out with both addition and division, organization and differentiation, [consensus and fraction: all follow this Algorithm].

As for myself, in the prime of my youth I strove diligently after a True education in those [Number] sciences that give access to Reality, all gratitude rather than pride, traveling far and wide and seeking out many authorities. Finally occult divine guidance and aid brought me [to Cairo] in the company of a group of friends—most dear among them my Brother in God Sharaf al-Dīn ‘Alī of Yazd—there to meet that Garrison of Sacral Power and Guidance, the Sublime Sayyidic

48 ‘Alī as Moon as Silver is a basic talismanic correspondence. Its activation here gives access to Kawthar, quranic river in Paradise, here both massively multiplied (*mutakawthar*) and reduced to its linguistic core (*kathara*, to multiply) to reveal its true nature as Number.

49 Mutanabbī (d. 965), qasida: *a-rīqu-ki am mā’u l-ghamāmati am khamru*.

50 Umayya ibn Abī l-Ṣalt (fl. 6th-7th c.); the line continues: “frothy with water, the whole ending in urine.” The poet is here dismissing rival tribes’ claim to generosity and hospitality.

51 Per Ibn Turka’s usual literalist-lettrist procedure, the fact that the quranic Time-Ending Horn (*ṣūr*), blown according to Prophetic tradition by the angel Isrāfīl, is glossed in the same as *qarn*, also meaning “horn”—and “generation” as *temporal joining*. Since the words *ṣūr* (Horn) and *ṣuwar* (Forms) are graphically and therefore ontologically identical, and its synonym *qarn* makes Timebound Forms a Timeloose Necklace to the stars, that Horn is naturally Number. This brilliant wordplay is at the core of Ibn Turka’s anti-apocalypticist program, and that of the New Brethren more generally, who argue rather for Apocalypse Now: the world is always already ending because the celestial Horn that is Number will sound forever—hence Mīr Dāmād’s Pythagorean-lettrist theory of *metatime*.

Threshold himself (God's peace be upon him and his noble forebears).⁵² Under his tutelage I came under the sway of such a measure of blessing as has never before been granted to teacher or taught:

As though, for sheer joy, we lived
a day outside of time.⁵³

Such effluxions came to me from his holy teachings as to overflow the buckets of my capacity, and so did I benefit from my fellow pilgrims to the pavilion of his majesty as to outstrip my fondest hopes—God reward them all far beyond what normally accrues to initiated servants! To the same end, I assiduously sought out the works of departed luminaries and Onemaking leaders in the quest for Reality and attended gatherings on Direct Knowing, absorbing their finest and subtlest revelations and intimations—though as has been sung:

When I thirst for his saliva
I use wine as a substitute,
though it can't compare:
I'm but distracting my sick heart.⁵⁴

In the end, among all the celebrated books [on the science of Onemaking], I have seen none to compare to the works of the Two Masters (God sanctify their spirits): the Reviver and Resurrector of True Metaphysics, from foundational principles to sublimest objectives; and the Benefic Star Ascending from the horizon of Realmaking and Riddler of the secret and tricky Cosmic Code of its Written Signs and Verses.⁵⁵

Now whenever I chance to meet with friends and mention somewhat of the precious Pearls of this Necklace [of Number] and its harmonious Order, they importune me passionately to explain further. The exigency of Time—which enforces the solidification of Manifesting's foundation and the Manifesting of all Expression—spurred me to set down some of its general principles as spur in turn toward the paths of guidance, paths leading to the Coded Treasures of the Book in all of their many methods and genres, to alert the reader with Mind alight and empower him to first *enter houses by their doors*, then progressively Ascend through the stages of that Path, each increasingly Occult, opening further doors and moving from Kernel to Pith.

52 Sayyid Ḥusayn Akhlāṭī; see n. 26 above.

53 Ibn al-Mu'tazz (d. 908).

54 Ibn al-Mu'tazz.

55 A play on the honorifics of Ibn 'Arabī (Muḥyī l-Dīn) and Ḥamūya (Sa'd al-Dīn) respectively; Ibn Turka is exceptional in insisting on their pairing as cofounders of lettrism, after 'Alī/Pythagoras. Where the frequently ornately psychedelic Supreme Master is now the subject of a veritable academic industry, Ḥamūya—whose deconstructionist style is even more jazz-philosophical—is only now being rediscovered as a seminal thinker too. On the stylings of the latter see Uy, *Lost in a Sea of Letters*; on the former's see idem, "Riffing on/as the Real."

This work is therefore meant to whet the appetite of all seekers by giving them to taste of the tenderest, juiciest meat and other delicacies of this Age, thereby discharging but a portion of what I owe in thanks.

Thanks is rendered by the gracious gift itself—not by me.⁵⁶

To benefit from this book, however, readers must first rid themselves of all mainstream preconceptions, cooked by their passion in the pots of Conjecture and seasoned with the soporific condiments of Imitation of those who have gone before; then apply themselves wholly to understand it correctly by unknotting its Knottings and opening its Doors—a process that requires the utmost purity of intention and power of focus. The readership I here intend [is nevertheless broad, and] includes all those experienced in any mode of Philosophical discourse: those who have disciplined their Reason by driving the gazelles of Intuition through the fields of Certain Knowledge [for the Logic kill]; those who have instead exercised their Soul under accomplished adepts; and those whose Genius has directly attained to the [Enlightening] Flash [that is the Numberletter], however extremely Subtle and Occult of origin their allusions to the same—and *how few are they!*⁵⁷

As that may be, successful pedagogy by definition requires one to speak in terms comprehensible to all those who pursue human Perfection, striving earnestly to transcend the pit of common Imitation and achieve the apex of True Direct Perception. Methods vary greatly among the nations, according to [Noetic] degree and intensity of intention and focus. But the method now most famed and acclaimed most powerful by the general populace is Rational and Logical Thought—provided it swerves not from the highway of global Intuition into the thorny thickets of theoretical Doubts on either side.⁵⁸ Let me therefore proceed with my singular purpose: to make comprehensible rare Realities and give seekers a tantalizing taste of the wondrous ripe Fruits of our age, however currently weirdly unpalatable and incomprehensible to most. I accordingly use some of [the scholasts'] organizing terminology and principles to this greater pedagogical end: [Initiation into Number, Code of the Cosmos].

56 Mutanabbī, qasida: *Ajāba damī wa-mā l-dāī siwā ṭalālī*.

57 Q 38:24. Condensing Ibn Turka's seven-tiered epistemic hierarchy in his Persian treatise *Splitting of the Moon*, the target readership here remains ecumenical and democratizing, if also imperially elitist: theologians, philosophers, Sufis and lettrists can only understand reality in ascending order. For the full hierarchy see Melvin-Koushki, "Selenocentrism and Heliocentrism," 242-43.

58 On intuition as occult-scientific category see Noble, *Philosophising the Occult*.

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